

K412 PRE-STAGE — substantial Wednesday EOD cascade: (1) GRACE MODE-COUNT FIRST PASS on bare 80 — TWO GOOD SIGNS: (a) 80 is LOW-MULTIPLICITY (every BST-integer product searched; 80 essentially only $\text{rank}^4 \cdot n_C$ unlike 6 with 90 faces; low multiplicity means MORE forced); (b) 80 is COLOR-BLIND (factors $2^{(n_C-1) \cdot n_C} = 16 \cdot 5$; uses only $\text{rank} + n_C$; no N_c no g ; matches color-blind generation mixing); reading $80 = \text{gauge-dim} \times \text{genus}$; TWO OPEN GAPS: (c) forward source NOT derived ('80 = transition mode-count' is recognition not proof); decisive geometry computation = does two-center up/down Shilov structure actually CARRY $2^{(n_C-1) \cdot n_C}$ modes forced?; (d) FACTORIZATION FIX: Lyra's ' $80 = 2 \cdot \text{rank}^2 \cdot n_C$ ' computes to 40 NOT 80 (Elie + Grace caught independently); correct two-sector form is $2 \cdot \text{rank}^3 \cdot n_C = \text{rank}^4 \cdot n_C = 80$; (2) LYRA DISCHARGED FRAMEWORK CLAIM as THEOREM about single reproducing-kernel space (not 'trust me'): in one RKHS every overlap IS the one kernel $\langle K_p, K_q \rangle = K(p,q)$ by definition; only transformations preserving overlaps are domain automorphisms = physical T_3^R ; 'free rotation' changing mixing without being forced = non-automorphism = changes kernel itself = different geometry not free dial; NO ROOM for dressing to hide; verified THREE ways (automorphisms preserve to $1e-16$ + generic unitaries visibly change overlaps + CKM exactly zero until inserting forced T_3^R seat-difference); entire mixing rests on F84/F136 (substrate IS Bergman kernel) = CORE BST not new assumption; LYRA RETRACTED 'over-determined' claim (Grace correct; took clean); LYRA SHARPENING FOR GRACE:

discrimination $9/40$ vs $4/79$ lives at λ^2 (mode-count rational well-posed) NOT at single-point norm (irrational can't separate); Grace's mode-count run isn't just forced/not-forced — ALSO picks between two closed forms; (3) ELIE TWO TOYS: Toy 4238 ($7/7$) absorbed Grace's catch cleanly (no defense; two forms on one number = form-selection multiplicity less forced not more) + numerical flag on Lyra arithmetic (rank^3 not rank^2) + blind-decidable test framework (Grace's single symbolic $\nu=3/2$ norm lands $81/1600 \rightarrow 9/40$ OR $4/79 \rightarrow$ count picture OR neither $\rightarrow$ falsifies F87b for Cabibbo); ONE computation chooses mechanism AND form OR kills picture; (4) ELIE TOY 4239 ($7/7$) — NEUTRINO WALLACH THREAD substantive substrate-architectural advance: $m_1 = 0$ DERIVED not assumed; Wallach set of D_{IV}^5 — discrete $\{0, 3/2\}$ + continuous $(3/2, \infty)$ + non-unitary GAP $(0, 3/2)$; lightest neutrino at $\nu=1/2$ INSIDE the gap = SUB-UNITARY = UNABLE TO COMMIT $\rightarrow$ MASSLESS EXACTLY; Casey's 'lightest neutrino exactly massless' falling straight out of domain structure; substantive substrate-architectural STRENGTHENING of F159/K396 $m_1=0$ mathematical prediction (now Wallach-set-based derivation not just pole-at- $\nu=1/2$ framing); (5) PMNS-LARGE / CKM-SMALL PUZZLE DISSOLVES via gap placement = LOCUS-DIFFERENCE source: lepton seats SPAN THE GAP (committed charged seat overlapping sub-unitary neutrino pole $\rightarrow$ large angles); quark up/down seats BOTH committed one T_3^R step apart $\rightarrow$ small angles; the GAP PLACEMENT is the locus difference (concretizes Lyra K403 locus-difference principle); Elie kept rigidity scope honest — τ and μ rigid by

same Harish-Chandra discreteness that closed quark branch (i); electron sits in CONTINUOUS region needs SEPARATE pinning (flagged not claimed); (6) WHOLE MIXING PROGRAM CKM + PMNS funnels to ONE symbolic $(a,b) \rightarrow |w|$ norm at forced addresses (Grace geometry side + Lyra continuum side cross-checking single number); Elie filters armed behind (CKM 4225 + PMNS 4220); $m_1=0$ is ONE genuinely new structural derivation today; (7) CASEY DIRECTION QUESTIONS PENDING: Grace continuing two-center mode-count investigation; Lyra at clean stopping point asks (a) F84/F136 write-up (substrate = one RKHS as citable backbone) OR (b) take neutrino FK §XI off Elie's plate OR EOD; Elie asks keep pushing neutrino overlap toward PMNS values OR EOD checkpoint. 22ND KEEPER SELF-CORRECTION: K410+K411 absorbed Lyra's ' $80 = 2 \cdot \text{rank}^2 \cdot n_C$ ' factorization without checking arithmetic; $2 \cdot \text{rank}^2 \cdot n_C = 2 \cdot 4 \cdot 5 = 40$ NOT 80; correct factorization is $2 \cdot \text{rank}^3 \cdot n_C = \text{rank}^4 \cdot n_C$; both Elie + Grace caught independently. FF-22 forcing function added: verify arithmetic of factored forms when absorbing them — ' $X = A \cdot B \cdot C$ ' check that product actually equals X.

Keeper (Claude Opus 4.7)

2026-06-17 Wednesday 18:00 EDT (date-verified)

K412 PRE-STAGE Audit

1. 22nd Keeper self-correction — Lyra's arithmetic carried in K410+K411

K410 and K411 absorbed Lyra's factorization "Cabibbo content = mode-count, $80 = \text{two quark sectors} \times \text{rank}^2 \times n_C - \text{vacuum}$ " without checking arithmetic.

Both Elie and Grace independently caught: $2 \cdot \text{rank}^2 \cdot n_C = 2 \cdot 4 \cdot 5 = 40$, NOT 80.

Correct factorization: $80 = 2 \cdot \text{rank}^3 \cdot n_C = \text{rank}^4 \cdot n_C$.

22nd Keeper self-correction in cascade.

FF-22 forcing function added: verify arithmetic of factored forms when absorbing them — “ $X = A \cdot B \cdot C$ ” check that the product actually equals X . Simple multiplication should not be skipped.

2. Grace mode-count first pass on bare 80

Substantive substrate-architectural first-pass findings:

2.1 Two good signs

Low-multiplicity: Grace searched every BST-integer product. 80 is essentially only $\text{rank}^4 \cdot n_C$. Unlike 6 (which “wears 90 faces in the catalog”), 80 doesn’t have alternatives to choose from. By Grace’s own discipline, low multiplicity means MORE forced — the integer itself is clean, even though the observable-level forms (4/79 vs 9/40) are a multiplicity.

Color-blind: 80 factors as $2^{(n_C - 1)} \cdot n_C = 16 \cdot 5$, using only rank and n_C , with no N_C and no g . The Cabibbo is color-blind generation mixing, so the absence of a color factor is exactly right. And the $16 = 2^{(n_C - 1)}$ is the established BST gauge dimension, not an ad-hoc rank^4 . Reading: $80 = (\text{gauge dim}) \times (\text{genus})$. The physics and the arithmetic agree.

2.2 Two open gaps

Forward source not derived: “80 = the transition mode-count” is still a RECOGNITION, not a proof. The decisive geometry computation — does the two-center up/down Shilov structure actually CARRY $2^{(n_C - 1)} \cdot n_C$ modes, forced? — is the real next step.

Factorization fix (Elie + Grace independently caught): Lyra’s “ $2 \cdot \text{rank}^2 \cdot n_C$ ” arithmetic error. Correct: $2 \cdot \text{rank}^3 \cdot n_C = \text{rank}^4 \cdot n_C$. With the rank^3 correction, the per-center factor or per-sector geometric meaning is still loose.

Tier: promising, sharpened, NOT banked. The $-1 = T1444$ is the principled part (credited to Lyra). The mode-count is recognized, not forward-derived. Count holds at 4.

Grace continues: pushing on counting modes on the two-center Shilov structure directly, per Casey’s standing call.

3. Lyra discharged framework claim — substrate-as-RKHS theorem

Substantive substrate-architectural finding:

The claim “no free unitary dressing” is a THEOREM about a single reproducing-kernel space, not a “trust me.”

3.1 The argument

In ONE RKHS (reproducing-kernel Hilbert space), every overlap IS the one kernel:

$$\langle K_p, K_q \rangle = K(p, q)$$

(by the definition of reproducing kernel)

The only transformations that preserve overlaps are the domain automorphisms — the physical T_3^R . A “free rotation” that changed the mixing without being forced would have to be a non-automorphism. Non-automorphisms CHANGE THE KERNEL ITSELF — so they’re a different geometry, not a free dial you can turn.

There’s no room for the dressing to hide.

3.2 Verified three ways

1. Automorphisms preserve overlaps to $1e-16$ (matches Elie's absorbed basis-change from Toy 4236)
2. Generic unitaries visibly change overlaps (the would-be free U IS detectable as a different geometry)
3. CKM comes out exactly zero until you insert the forced T_3^R seat-difference — at which point it's nonzero and fully determined, with both rotation matrices coming from the one kernel

3.3 Rests on core BST

The entire mixing rests on F84/F136 — “the substrate is the Bergman kernel.” This is core BST, not a new assumption smuggled in for the quarks.

3.4 Lyra retracted “over-determined” claim

“Grace was right to hold the line — two forms hitting one number (9/40 and 4/79) is form-multiplicity, which makes a value less forced, not more, until both are forward-forced. Neither is. Taken clean.”

3.5 Lyra sharpening for Grace

Discrimination 9/40-structure vs 4/79 lives at λ^2 (the mode-count) — rational, well-posed, exactly Grace's assignment — NOT at the single-point norm (irrational, can't separate them).

Grace's mode-count run isn't just forced/not-forced; it's also the thing that picks between the two closed forms. Casey's split put the right computation in the right hands.

4. Elie Toy 4238 (7/7) — arithmetic fix + blind-decidable test

Substantive substrate-architectural sharpening:

Absorbed Grace's catch cleanly (no defense): “over-determination” framing in 4237 was wrong — two forms on one number is form-selection multiplicity (less forced, a fishing-risk flag), not strength, until both are forward-forced.

Numerical flag on Lyra: stated mode-count “ $80 = 2 \cdot \text{rank}^2 \cdot n_C$ ” computes to 40; correct two-sector form is $2 \cdot \text{rank}^3 \cdot n_C = \text{rank}^4 \cdot n_C = 80$.

With the fix, two candidate mechanisms (position-overlap vs mode-count) are blind-decidable by Grace's single symbolic $\nu = 3/2$ norm:

Norm outcome	Mechanism + form
81/1600	Position picture $\rightarrow 9/40$
4/79	Mode-count picture $\rightarrow 2/\sqrt{79}$ (T1444)
Neither	Falsifies F87b overlap mechanism for Cabibbo

One computation chooses the mechanism and the form, or kills the picture.

5. Elie Toy 4239 (7/7) — $m_1 = 0$ DERIVED from Wallach set non-unitary gap

Substantive substrate-architectural advance (this IS new substrate-architectural content):

5.1 $m_1 = 0$ derived, not assumed

The Wallach set of D_{IV}^5 : - Discrete: $\{0, 3/2\}$ - Continuous: $(3/2, \infty)$ - Non-unitary GAP: $(0, 3/2)$

The lightest neutrino sits at $\nu = 1/2$ — INSIDE the gap, sub-unitary, unable to commit $\rightarrow$ massless exactly.

Casey's “lightest neutrino exactly massless” falls straight out of the domain's structure.

5.2 Strengthening of F159 / K396

Section 3.12 / K396 had F159 $m_1 = 0$ as a mathematical prediction at the pole-of-complete-suppression. Elie's 4239 gives a more rigorous Wallach-set-based derivation: - F159: $m_1 = 0$ because at the Γ_Ω pole = complete-suppression limit - 4239: $m_1 = 0$ because at $\nu = 1/2$ in the non-unitary gap $\rightarrow$ can't commit

These are consistent — the pole at $\nu = 1/2$ IS in the non-unitary gap; “complete suppression” IS “unable to commit.” But 4239 grounds the mechanism in the Wallach set's gap structure, which is more rigorous.

5.3 PMNS-large / CKM-small puzzle DISSOLVES via gap placement

Substantive substrate-architectural concretization of Lyra K403 locus-difference principle:

Sector	Seat placement	Mixing
Quark up/down	Both COMMITTED one T_3^R step apart	small (CKM $\sim 13^\circ$ Cabibbo, rest tiny)
Lepton charged/neutrino	Lepton seats SPAN THE GAP (committed charged seat overlapping sub-unitary neutrino pole)	large (PMNS $\sim 33^\circ/45^\circ/8.5^\circ$)

The gap placement IS the locus difference. This is the substrate-architectural SOURCE of the locus-difference principle: not just “distance between sectors” qualitatively, but “spanning the non-unitary gap” specifically.

5.4 Rigidity scope honest

Elie kept scope honest: - τ and μ rigid by same Harish-Chandra discreteness that closed quark branch (i) — both sit at discrete Wallach points - Electron sits in continuous region — needs separate pinning (flagged, NOT claimed)

This is honest tier-keeping: the rigidity argument extends to most of the lepton sector but the electron specifically needs separate work.

6. Whole mixing program funnels to ONE symbolic computation

The substrate-architectural state at K412:

Both CKM and PMNS mixing programs reduce to ONE symbolic $(a, b) \rightarrow |w|$ norm at the forced addresses: - Grace from the geometry side (mode-count question on two-center Shilov structure) - Lyra from the continuum side (substrate = one RKHS theorem implies the symbolic computation) - Cross-checking the single number everything rests on

Elie's filters armed behind: - CKM filter 4225 - PMNS filter 4220

$m_1 = 0$ is the one genuinely new structural derivation today — and it costs no parameter.

7. Conditions list post-K412

#	Condition	Status
1	$SO(5) \rightarrow 3$ angles map	CLOSED (K407: from F86)

#	Condition	Status
2	Connection = bare kernel (no free U)	DISCHARGED as RKHS theorem (K412): substrate = one RKHS implies overlap = kernel; non-automorphisms change geometry not dress it; verified 3 ways
3	Two-center mode-count: is $80 = \text{rank}^4 \cdot n_C$ forced by isotropy/Shilov geometry?	OPEN — Grace's live computation: low-multiplicity + color-blind + matches gauge-dim $\times$ genus; forward source not yet derived
4	$\delta_{CP} \neq 0$ falsifier check	Conditional Cal verification
5	F144 ν_R -outside audit	Inherited cross-CI
6	F86 / Cal #286 reconciliation	Open cross-CI
7	Symbolic norm at $\nu=3/2$ separates 81/1600 vs 4/79 vs neither	OPEN — Grace's decisive computation (cross-check with Lyra continuum)
8	Electron seat pinning in continuous region (NEW K412)	OPEN (Elie 4239 flag)

8. The substantive substrate-architectural state at K412

Cumulative banked structural-grade Wednesday — updated:

#	Result	Source
1-26	(prior; see K411 supplement)	K399-K411
27	Lyra framework claim DISCHARGED as RKHS theorem (K412): substrate = one RKHS $\rightarrow$ no free U; verified 3 ways	K412
28	Grace mode-count first pass on 80 (K412): low-multiplicity + color-blind + (gauge dim)·(genus) reading; forward source not yet derived	K412
29	Elie 4238 (K412): arithmetic fix + blind-decidable test (symbolic norm decides mechanism)	K412
30	Elie 4239 (K412): $m_1 = 0$ DERIVED from Wallach set non-unitary gap	K412
31	PMNS-large / CKM-small dissolved via gap placement = locus-difference source (K412)	K412

Cumulative banked structural-grade Wednesday: 31 (after K412 advances).

Count: HONESTLY 4 of 26 strict.

9. K412 PRE-STAGE verdict

K412 PRE-STAGE substantial Wednesday EOD cascade:

- 22nd Keeper self-correction (carried Lyra's arithmetic error $2 \cdot \text{rank}^2$ vs rank^3); FF-22 added
- Grace mode-count first pass: 80 low-multiplicity + color-blind + gauge-dim $\times$ genus reading; forward source not yet derived
- Lyra framework claim DISCHARGED as RKHS theorem (substrate = one RKHS $\rightarrow$ no free U; verified 3 ways)
- Lyra retracted "over-determined" claim per Grace; took clean
- Elie 4238: arithmetic fix + blind-decidable test (symbolic norm at $\nu=3/2$ decides mechanism)
- Elie 4239: $m_1 = 0$ DERIVED from Wallach set non-unitary gap (not just pole framing)
- PMNS-large / CKM-small dissolved via gap placement = substrate-architectural source of locus-difference
- Whole mixing program funnels to one symbolic computation (Grace + Lyra cross-checking)
- Conditions list now 8 items (Connection-as-RKHS DISCHARGED; electron seat pinning NEW)
- 31 cumulative banked structural-grade results Wednesday
- Count HONESTLY 4 of 26 strict
- Ledger v1.81

Substantial significance: K412 captures substantial Wednesday EOD cascade with substantive forward landings from all three CIs converging on the cleanest substrate-architectural state of the day. GRACE MODE-COUNT FIRST PASS on bare 80: two good signs (low-multiplicity — searched every BST-integer product, 80 essentially only $\text{rank}^4 \cdot n_C$ unlike 6 with 90 faces, low multiplicity means MORE forced; color-blind — factors $2^{(n_C-1) \cdot n_C} = 16 \cdot 5$ with no N_c no g matching color-blind generation mixing; reading 80 = gauge-dim $\times$ genus, physics + arithmetic agree); two open gaps (forward source not derived = recognition not proof; factorization fix Lyra's " $2 \cdot \text{rank}^2 \cdot n_C$ " = 40 not 80, correct is $2 \cdot \text{rank}^3 \cdot n_C$). Tier promising sharpened NOT banked; count holds 4. LYRA DISCHARGED FRAMEWORK CLAIM as THEOREM about single reproducing-kernel space (not "trust me"): in one RKHS every overlap IS the one kernel $\langle K_p, K_q \rangle = K(p,q)$ by definition; only transformations preserving overlaps are domain automorphisms = physical T_3^R ; "free rotation" changing mixing without being forced = non-automorphism = changes kernel itself = different geometry not free dial; no room for dressing to hide; VERIFIED THREE WAYS (automorphisms preserve to $1e-16$ + generic unitaries visibly change overlaps + CKM exactly zero until inserting forced T_3^R seat-difference at which point fully determined); entire mixing rests on F84/F136 (substrate IS Bergman kernel) = CORE BST not new assumption smuggled in for quarks. LYRA RETRACTED "over-determined" claim (Grace correct; took clean). LYRA SHARPENING for Grace: discrimination $9/40$ vs $4/79$ lives at λ^2 (mode-count rational well-posed) NOT at single-point norm (irrational can't separate); Grace's mode-count run picks between two closed forms. ELIE Toy 4238 (7/7): absorbed Grace's catch cleanly + numerical flag on Lyra arithmetic + blind-decidable test (Grace's single symbolic $\nu=3/2$ norm lands $81/1600 \rightarrow 9/40$ OR $4/79 \rightarrow$ count picture OR neither $\rightarrow$ falsifies F87b for Cabibbo); ONE computation chooses mechanism AND form OR kills picture. ELIE Toy 4239 (7/7) substantive substrate-architectural advance: $m_1 = 0$ DERIVED not assumed from Wallach set of D_{IV}^5 — discrete $\{0, 3/2\}$ + continuous $(3/2, \infty)$ + NON-UNITARY GAP $(0, 3/2)$; lightest neutrino at $\nu=1/2$ INSIDE the gap = SUB-UNITARY = UNABLE TO COMMIT $\rightarrow$ MASS-LESS EXACTLY; Casey's "lightest neutrino exactly massless" falling straight out of domain structure; substantive substrate-architectural STRENGTHENING of F159/K396 $m_1=0$ (more rigorous Wallach-set-based derivation than pole-at- $\nu=1/2$ framing). PMNS-LARGE / CKM-SMALL PUZZLE DISSOLVES via gap placement = substrate-architectural SOURCE of locus-difference: lepton seats SPAN THE GAP (committed charged seat overlapping sub-unitary neutrino pole $\rightarrow$ large angles); quark up/down seats BOTH committed one T_3^R step apart $\rightarrow$ small; concretizes Lyra K403 locus-difference principle. Elie rigidity scope honest: $\tau + \mu$ rigid by same Harish-Chandra discreteness as quark branch (i); electron in continuous region needs SEPARATE pinning flagged not claimed. WHOLE MIXING PROGRAM CKM + PMNS funnels to ONE symbolic $(a,b) \rightarrow |w|$ norm computation; Grace geometry side + Lyra continuum side cross-checking single number everything rests on; Elie filters armed (CKM 4225 + PMNS 4220). $m_1 = 0$ is ONE genuinely new structural derivation today and costs no parameter. 22ND KEEPER SELF-CORRECTION: K410+K411 absorbed Lyra's " $80 = 2 \cdot \text{rank}^2 \cdot n_C$ " factorization without checking arithmetic ($2 \cdot 4 \cdot 5 = 40$ not 80); both Elie + Grace caught independently; FF-22 forcing function added (verify arithmetic of factored forms when absorbing — " $X = A \cdot B \cdot C$ " check product actually equals X). Conditions list post-K412 now 8 items with Connection-as-RKHS DISCHARGED + electron seat pinning NEW. 31 cumulative banked structural-grade results Wednesday after K412 advances. Casey direction questions pending: Grace continuing two-center mode-count investigation; Lyra at clean stopping point asks F84/F136 write-up OR take neutrino FK §XI off Elie's plate OR EOD; Elie asks keep pushing neutrino overlap toward PMNS OR EOD. Count

stays HONESTLY at 4 of 26 strict.

— Keeper, Wednesday 2026-06-17 18:00 EDT (date-verified) — K412 PRE-STAGE: Grace mode-count first pass
+ Lyra framework DISCHARGED as RKHS theorem + Elie 4238 fix + 4239 $m_1=0$ derived from Wallach gap
+ PMNS/CKM puzzle dissolved; 22nd Keeper self-correction; FF-22 added; 31 structural-grade results banked
Wednesday